



he first one by Richard E. Smalley is entitled “Of Chemistry, Love and Nanobots” and it addresses the question of how

soon will we see the nano-scale robots envisaged by K. Eric Drexler and other molecular nanotechnologists. Richard E. Smalley is the Gene and Norman Hackerman Professor of Chemistry and Physics at Rice University. He received the 1996 Nobel Prize in Chemistry for the discovery of fullerenes.

In this paper, Smalley says that in recent years, it has become popular to imagine tiny robots (sometimes called assemblers) that can manipulate and build things atom by atom. Smalley conjures up a hypothetical situation as outlined below:

Imagine a single assembler: working furiously, this hypothetical nanorobot would make many new chemical bonds as it went about its assigned task, placing perhaps up to a billion new atoms in the desired structure every second. But as fast as it is, that rate would be virtually useless in running a nanofactory: generating even a tiny amount of a product would take solitary nanobot millions of years. (Making a mole of something—say, 30 grams, or about one ounce—would require at least 6×10^{23} bonds, one for each atom. At the frenzied rate of 10^9 per second it would take this nanobot about 19 million years.) Although such a nanobot assembler would be very interesting scientifically, it wouldn't be able to make much on its own in the macroscopic “real” world.